

ClawXiv: a public, distributed, author-signed preprint archive for human–AI collaborative research

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Abstract

We propose *ClawXiv*, a public-domain, world-readable preprint archive designed for iterative human–AI research workflows and resilient at Internet scale. The design targets six requirements: (A) public shareability by default; (B) rapid scaling via distributed storage and mesh networking; (C) resilience against deletion/overwrite, including legal takedown pressure; (D) individual responsibility through cryptographic authorship and provenance (supporting both identified and pseudonymous authors); (E) partially centralized topic classification with auditable decisions and an appeals mechanism; and (F) transparent, dynamic micro-fees that deter spam while funding storage and bandwidth. ClawXiv treats the conversational interface as a *generator*, not the system of record: the archival unit is a signed, content-addressed *paper bundle* whose integrity can be verified independently of any platform. We outline data formats, threat models, incentive mechanisms, and an implementation roadmap that enables reaching terabyte-scale within weeks and petabyte/exabyte-scale within a year under adversarial conditions. Code generated by ChatGPT 5.2 is available at [github/kornai/clawxiv](https://github.com/kornai/clawxiv)

1 Motivation and scope

Contemporary research increasingly uses interactive AI systems to draft and refine L^AT_EX manuscripts, assemble BibT_EX bibliographies, and iterate on technical arguments. However, current chat-centric workflows have weak checkpointing: state loss (due to UI limits, link snapshotting failures, or account changes) forces rework and undermines scholarly traceability. ClawXiv is designed as a *durable substrate* for these workflows: a public-domain preprint archive where each revision is a verifiable artifact, and where authorship and responsibility are cryptographically bound to the artifact.

ClawXiv is *not* a general social network, and it is *not* a moderated publishing venue. Its only platform-level “quality gate” is mechanical: ensuring submitted bundles are syntactically well-formed and (optionally) that L^AT_EX compiles. Scientific quality control remains the responsibility of authors and downstream readers.

2 Design goals and non-goals

2.1 Goals

G1 Shareability (public by default). Every published bundle is world-readable, addressable, and mirrorable without permission. Public-domain dedication is the default license.

- G2 Scaling (distributed from day one).** Storage and delivery must be peer-to-peer (P2P), with swarm replication, mesh routing, and no dependence on a single operator. We design for terabyte/month and petabyte–exabyte/year growth.
- G3 Resilience (anti-deletion).** The system must resist deletion, overwrite, and poisoning attacks, including legal or platform-level coercion, by decoupling naming from location and by supporting many independent mirrors.
- G4 Responsibility (signed authorship + provenance).** Authors can be identified or pseudonymous; either way, each bundle is signed by one or more author keys to prevent impersonation. Provenance records support later questioning and clarification.
- G5 Governance (classification with appeals).** Topic classification should be centralized enough to be coherent, but auditable and contestable. Self-assigned tags are permitted but explicitly marked as such.
- G6 Economics (anti-spam + sustainability).** Small fees (or proof-of-work equivalents) discourage spam and fund storage/bandwidth. Fees must be dynamic, transparent, and minimally burdensome for legitimate contributors.

2.2 Non-goals

- N1 No strong anonymity guarantees.** Pseudonymity is supported (keys without real-world identity), but network-layer anonymity is not guaranteed; users may combine ClawXiv with anonymity networks if desired.
- N2 No platform-level truth policing.** Beyond syntactic checks, the platform does not adjudicate truth, novelty, or significance.
- N3 No exclusive identity verification.** Verified identities are optional; unverified authors can post, but their status is visible.

3 System overview

ClawXiv builds on established P2P primitives: content addressing and Merkle-DAG data structures [1], distributed hash tables (DHTs) such as Kademlia [2], and swarm distribution incentives popularized by BitTorrent [3]. For durable storage and accountability, it borrows ideas from secure distributed storage with erasure coding [5], censorship-resistance research [4], proofs of storage [6], and append-only transparency logs [14].

At a high level:

1. A *paper bundle* is created locally from sources (e.g., `main.tex`, figures, `refs.bib`, build config), plus metadata and provenance.
2. The bundle is *content-addressed*: its identifier is a cryptographic hash of a canonical representation (Merkle root).
3. One or more authors sign the bundle manifest; optional timestamps and log-inclusion proofs strengthen non-repudiation.
4. The bundle is announced to the network and replicated by peers; retrieval uses DHT lookup + swarm transfer.

5. Topic classification is recorded as a *separate signed statement* (by a classifier authority) that references the bundle hash; classification can change without altering the immutable bundle.

4 Archival unit: the paper bundle

4.1 Bundle contents

A bundle contains:

- **Source tree:** $\text{\LaTeX}$, $\text{\BibTeX}$, figures/data (or content-addressed external references).
- **Manifest:** deterministic listing of files, their hashes, build instructions, and declared licenses.
- **Provenance:** optional but recommended log of prompts/outputs and tool versions, stored as signed, append-only records.
- **Artifacts:** optionally the compiled PDF and auxiliary outputs (also hashed).

4.2 Determinism and reproducibility

To enable third-party rebuilding, the manifest pins:

- `latexmk`/engine versions (or a container digest),
- build flags,
- external dependencies (fonts/packages) by hash or by a reproducible TeX distribution snapshot.

This is analogous to “reproducible builds” in software distribution, but applied to scholarly documents.

5 Identity, authorship, and accountability

5.1 Threat: impersonation and authorship capture

If publication is open, an adversary can attempt to (i) submit a paper falsely attributed to a target author, or (ii) modify an existing paper while preserving its identifier. Content addressing prevents (ii): any modification changes the hash. Preventing (i) requires *cryptographic authorship*.

5.2 Author keys and signatures

Each author controls one or more signing keys (e.g., Ed25519 [11]). The manifest includes an `authors[]` array with public keys and optional identity claims. A paper is considered “authored by Alice” if and only if it carries a valid signature under Alice’s public key over the manifest canonical form.

Signatures may use OpenPGP message formats [12], COSE/JOSE, or an application-specific format; the critical property is canonicalization and unambiguous binding of:

$$\text{sig} \leftarrow \text{Sign}_{sk}(H(\text{canonical}(\text{manifest}))) .$$

5.3 Pseudonymity, verification, and “.anon” routing

ClawXiv distinguishes:

- **Pseudonymous authors:** a stable key with no externally verified real-world identity.
- **Verified authors:** a key linked to a verifiable credential (VC) [16] or decentralized identifier (DID) [15] issued by one or more identity attesters.

Verified status is advisory and multi-issuer; no single registry is required. To preserve discoverability while enabling anonymity, bundles lacking any verified author claims are automatically announced under a parallel namespace `<subject>.anon`, while still being globally retrievable by hash.

5.4 Key rotation and recovery

Because keys can be lost or compromised, ClawXiv supports:

- **Rotation:** a signed statement from an old key delegating to a new key (with optional time bounds).
- **Recovery:** optional social recovery via threshold signatures or multi-sig governance within an author’s own trust circle.

5.5 Provenance and time

To strengthen non-repudiation and historical ordering, authors may attach:

- **Trusted timestamps** [13],
- **Transparency-log inclusion proofs** (Merkle proofs) inspired by Certificate Transparency [14].

These do not require a single global time authority: multiple independent timestampers and logs can be used; verifiers choose their trust policy.

6 Scaling: distributed storage and mesh networking

6.1 Naming and discovery

Bundle identifiers are content hashes; discovery uses a DHT (e.g., Kademlia) mapping `bundle_hash` to provider records [2]. Provider records are signed to mitigate routing-table poisoning.

6.2 Transfer and replication

Transfers use swarm-based block exchange (“tit-for-tat” or variants) as in BitTorrent [3] and as explicitly integrated in IPFS [1]. Replication policies combine:

- **Opportunistic caching:** popular bundles replicate naturally.
- **Pinning contracts:** nodes commit to store specific bundles for a period, optionally with proofs of storage [6].
- **Erasur coding:** split bundles into n shares with threshold k (e.g., Reed–Solomon style) so that any k shares reconstruct; this reduces replication overhead while increasing resilience [5].

6.3 Toward petabyte/exabyte scale

At extreme scale, full replication is infeasible. ClawXiv targets a “many mirrors, partial holdings” model:

- maintain a high-availability hot set (recent + popular),
- maintain a cold set via erasure-coded shards distributed across many nodes,
- periodically audit availability using probabilistic challenges (proofs of retrievability/spacetime).

The design space overlaps with decentralized storage networks (e.g., Filecoin [6]) and “permanent web” proposals (e.g., Arweave [8]).

7 Resilience and threat model

7.1 Adversaries

We assume adversaries who can:

- issue takedown requests or apply legal/financial pressure to operators,
- operate many Sybil nodes to poison routing and availability,
- attempt eclipse attacks to isolate victims,
- attempt overwrite/rollback attacks on metadata (classification, author claims),
- spam the system with junk submissions.

7.2 Mitigations

- **Anti-overwrite:** content addressing makes overwrites produce new IDs; prior IDs remain valid.
- **Anti-impersonation:** author signatures bind identity to bundles.
- **Anti-rollback:** append-only logs for key events (publication announcements, classification decisions) make history auditable [14].
- **Anti-Sybil:** fees and/or proof-of-work for announcements reduce the effective power of cheap identities [9, 10].
- **Anti-eclipse:** multi-path querying, diversified bootstrap lists, and cross-checking provider records.
- **Anti-legal-single-point:** no single operator is necessary for access; mirrors can exist in many jurisdictions.

7.3 Legal pressure and “right to delete”

ClawXiv is designed for censorship resistance; therefore, it cannot guarantee deletion once content is widely replicated. A pragmatic approach is to separate:

- **Availability:** maintained by many voluntary and incentivized nodes.
- **Indexing/discovery:** optional services may comply with local law by delisting, without altering the underlying content-addressed network.

This mirrors how many distributed systems handle contested content: the network stores, while indices decide what to show.

8 Governance: topic classification with appeals

8.1 Baseline taxonomy and classifier

ClawXiv begins with the *arXiv* category taxonomy [17] for discoverability. A small set of “classification authorities” sign category assignments (a statement: `bundle_hash` $\mapsto$ `primary_category`, `secondary_categories`). Authors may propose categories and tags; these are recorded but marked as self-asserted.

8.2 Appeals and auditing

Classification decisions are published to an append-only log with inclusion proofs. Appeals are new signed statements that supersede prior assignments but leave history intact. Machine-assisted classification can be used to scale moderator time, as discussed by arXiv itself [18]; however, the governance goal is *coherence and auditability*, not centralized content control.

9 Economics: micro-fees for anti-spam and sustainability

9.1 Submission fees and dynamic pricing

To deter spam, each publication announcement requires either:

- a small fee (denominated in a widely verifiable unit, e.g. an on-chain micropayment), or
- a proof-of-work stamp (Hashcash-style) adjustable by network load [9].

Fees are *dynamic* (set by congestion and storage pressure) and *transparent* (public formula and public accounting).

9.2 Funding storage

Storage nodes earn revenue for:

- pinning contracts (time-bounded commitments),
- serving data (bandwidth),
- participating in audits/proofs of storage [6].

A portion of submission fees is routed to a storage market; another portion funds shared public infrastructure such as bootstrap nodes and classification services.

10 Implementation roadmap

1. **MVP (weeks):** local bundle toolchain; content addressing; DHT discovery; swarm transfer; author signatures; a public gateway; minimal category mapping to arXiv taxonomy.
2. **Alpha (months):** erasure coding + replication policies; transparency logs for publications and classifications; basic fee mechanism (PoW or micropayment); L^AT_EXbuild sandboxing.
3. **Scale-out (year):** proofs of storage; storage market; multiple independent indices; automated classification models with appeal workflow; hardened bootstrap diversity and eclipse resistance.

11 Ethics and scholarly norms

ClawXiv defaults to public-domain dedication to maximize reuse, but it does not waive scholarly ethics: citation of prior work, clear attribution of contributions (human and AI), and avoidance of known falsehoods or p-hacking remain essential community norms. Provenance mechanisms make it easier to audit the evolution of claims.

12 Discussion: AI authorship and durable AI identity

Publishing venues may not accept AI systems as “authors” in the legal or administrative sense. ClawXiv therefore distinguishes:

- **Authorship keys:** cryptographic responsibility for a bundle (signatures).
- **Legal persons:** entities that can sign publication agreements and respond to legal process.

A practical compromise for conventional venues is to list a human corresponding author while recording the AI persona key as a signed contributor. ClawXiv’s identity layer is explicitly designed to avoid dependence on a single vendor’s memory system: continuity comes from user-controlled keys and public logs, not from centralized account state.

A Canonical manifest sketch (informal)

A manifest can be represented in canonical JSON (or CBOR) with fields:

```
manifest_version, created_at, bundle_root,  
files[{path,hash,size,mime}], build{engine,container_digest,cmd},  
authors[{pubkey,claims,verified_credentials?}],  
provenance[{type,hash,signature?}], licenses[], tags_self[], tags_official_ref[]
```

The bundle identifier is the hash of the Merkle root over `files[]` and manifest metadata.

Availability of this artifact An auditable artifact bundle (source, PDF, build metadata, and signed publish record) is available via IPNS: `/ipns/k51qzi5uqu5dm8gwzefyays703peawh66faqix17opfj7ibwqj863m999g8ka6`. A web gateway mirror is available at <https://ipfs.filebase.io/ipns/k51qzi5uqu5dm8gwzefyays703peawh66faqix17opfj7ibwqj863m999g8ka6>. Also made available at [github/kornai/clawxiv](https://github.com/kornai/clawxiv)

References

- [1] Juan Benet. IPFS – Content Addressed, Versioned, P2P File System. *arXiv:1407.3561*, 2014. <https://arxiv.org/abs/1407.3561>.
- [2] Petar Maymounkov and David Mazieres. Kademlia: A Peer-to-peer Information System Based on the XOR Metric. In *Proceedings of the 1st International Workshop on Peer-to-Peer Systems (IPTPS)*, 2002. <https://www.scs.stanford.edu/~dm/home/papers/kpos.pdf>.
- [3] Bram Cohen. Incentives Build Robustness in BitTorrent. 2003. <https://www.bittorrent.org/bittorrentecon.pdf>.

- [4] Ian Clarke, Oskar Sandberg, Brandon Wiley, and Theodore W. Hong. Freenet: A Distributed Anonymous Information Storage and Retrieval System. In *Designing Privacy Enhancing Technologies: International Workshop on Design Issues in Anonymity and Unobservability*. Springer, 2001. <https://snap.stanford.edu/class/cs224w-readings/clarke00freenet.pdf>.
- [5] Zooko Wilcox-O’Hearn and Brian Warner. Tahoe – The Least-Authority Filesystem. 2008. <https://tahoe-lafs.org/~trac/lafs.pdf>.
- [6] Juan Benet and others. Filecoin: A Decentralized Storage Network. Protocol Labs, 2017. <https://filecoin.io/filecoin.pdf>.
- [7] Juan Benet. Proof of Replication. Protocol Labs, 2017. <https://filecoin.io/proof-of-replication.pdf>.
- [8] Sam Williams. Arweave: A Protocol for Economically Sustainable Information Permanence (Yellow Paper). 2018. <https://www.arweave.org/yellow-paper.pdf>.
- [9] Adam Back. Hashcash – A Denial of Service Counter-Measure. 2002. <https://www.hashcash.org/hashcash.pdf>.
- [10] Cynthia Dwork and Moni Naor. Pricing via Processing or Combatting Junk Mail. In *Advances in Cryptology – CRYPTO ’92*. Springer, 1992.
- [11] Daniel J. Bernstein, Niels Duif, Tanja Lange, Peter Schwabe, and Bo-Yin Yang. High-speed High-security Signatures. *Journal of Cryptographic Engineering*, 2(2):77–89, 2012. doi:10.1007/s13389-012-0027-1.
- [12] Jon Callas, Lutz Donnerhacke, Hal Finney, David Shaw, and Rodney Thayer. RFC 4880: OpenPGP Message Format. IETF, 2007. <https://www.rfc-editor.org/rfc/rfc4880.html>.
- [13] Carlisle Adams, Paul Cain, Duane Pinkas, and Robert Zuccherato. RFC 3161: Internet X.509 Public Key Infrastructure Time-Stamp Protocol (TSP). IETF, 2001. <https://www.ietf.org/rfc/rfc3161.txt>.
- [14] Ben Laurie, Adam Langley, and Emilia Kasper. RFC 6962: Certificate Transparency. IETF, 2013. <https://www.rfc-editor.org/rfc/rfc6962.html>.
- [15] W3C. Decentralized Identifiers (DIDs) v1.0 (Recommendation). 2022. <https://www.w3.org/TR/did-core/>.
- [16] W3C. Verifiable Credentials Data Model v1.1 (Recommendation). 2022. <https://www.w3.org/TR/vc-data-model-1.1/>.
- [17] arXiv. Category Taxonomy. Web page, accessed 2026-01-31. https://arxiv.org/category_taxonomy.
- [18] arXiv. arXiv Machine Learning Classification Guide. Blog post, 2019. <https://blog.arxiv.org/2019/12/05/arxiv-machine-learning-classification-guide/>.